

PFAS COMPLIANCE REPORT

Report ID: PFAS-2026-0331-0004
Generated: 2026-03-31T23:32:39
Engine: KOMPOSOS-III v1.2.0

Executive Summary

Materials Screened	PFAS Detected	PFAS-Free	Risk Level
4	2	2	MODERATE

This portfolio contains 2 PFAS substance(s) requiring attention. 8 replacement candidate(s) have been identified and scored for compatibility. See Sections 2-3 for detailed findings and replacement recommendations.

Section 2: PFAS Screening Results

Material	Function	PFAS Substance	Category	Urgency	CAS Number
PVDF	cathode binder	Polyvinylidene fluoride	fluoropolymer	MODERATE	24937-79-9
PTFE	gasket seal	Polytetrafluoroethylene	fluoropolymer	MODERATE	9002-84-0

2 additional material(s) passed PFAS screening: NMC811, Cu foil

Section 3: Replacement Analysis

PVDF - PFAS-Free Alternatives

Replacement	Score	Performance	Processability	Cost	Availability	Verdict
CMC+SBR	0.84	0.75	0.85	0.90	0.95	VALIDATED
PAA	0.77	0.70	0.80	0.85	0.80	VALIDATED
Alginate	0.70	0.55	0.75	0.95	0.70	VALIDATED
PAN	0.64	0.60	0.50	0.80	0.70	CAUTION

Provenance for CMC+SBR:
performance_match: Bresser et al., Energy Environ. Sci. 2018; Li et al., J. Electrochem. Soc. 2020
processability: Dams & Heylen, PFAS restrictions impact, 2023
cost_factor: OECD, PFASs and alternatives, 2022
availability: Commercial availability assessment 2026

PTFE - PFAS-Free Alternatives

Replacement	Score	Performance	Processability	Cost	Availability	Verdict
EPDM	0.77	0.65	0.80	0.85	0.90	VALIDATED
NBR	0.75	0.55	0.80	0.90	0.95	VALIDATED
PDMS	0.73	0.60	0.75	0.80	0.90	VALIDATED
PEEK	0.60	0.80	0.40	0.30	0.70	CAUTION

Provenance for EPDM:
performance_match: Bresser et al., Energy Environ. Sci. 2018; Li et al., J. Electrochem. Soc. 2020
processability: Dams & Heylen, PFAS restrictions impact, 2023
cost_factor: OECD, PFASs and alternatives, 2022
availability: Commercial availability assessment 2026

Section 4: Regulatory Timeline

Jurisdiction	Regulation	Status	Effective Date	Days Remaining
Stockholm Convention	PFOS elimination	banned	2009-05-22	-6157 (PAST)
EU	PFOA + C9-C14 PFCA ban	banned	2020-07-04	-2096 (PAST)
US	EPA PFAS reporting	restricted	2024-01-01	-820 (PAST)
US	EPA MCL (drinking water)	restricted	2024-04-10	-720 (PAST)
EU	PFHxA restriction (C6 PFCAs)	proposed_ban	2026-10-01	184
EU	Universal PFAS restriction	proposed_ban	2027-06-01	427

Section 5: Action Plan

Priority 1: PLAN: Evaluate alternatives for proposed-ban substances (365 days)

Rationale: These substances face proposed restrictions. Begin alternative evaluation to avoid supply disruption.

Affected

Priority 2: VALIDATE: Run pilot tests on VALIDATED replacement candidates (90 days)

Rationale: Replacement candidates scored >= 0.7 with published provenance. Pilot testing recommended before full qualification.

Affected

Section 6: Methodology

Engine: KOMPOSOS-III Chemistry Engine v1.2.0

Approach: Registry lookup + heuristic substring detection

Scoring: Weighted composite: 40% performance + 20% processability + 20% cost + 20% availability

Reference Databases:

- PFAS Registry (35 curated substances with CAS numbers)
- ECHA SVHC candidate list
- EU REACH Annex XVII
- Stockholm Convention on POPs
- US EPA PFAS Strategic Roadmap

Validation Statistics:

- Pfas Registry Size: 35
- Replacement Candidates: 30
- Test Count: 81
- Pass Rate: 100%

Limitations and Caveats:

- Scores based on published literature data; pilot testing recommended
- Heuristic detection may produce false positives for non-PFAS fluorinated materials
- Regulatory dates based on current proposals; subject to legislative changes
- Report generated relative to reference date 2026-03-31

Section 7: Audit Certificate

Report ID	PFAS-2026-0331-0004
Generated	2026-03-31T23:32:39.989154Z
Engine Version	1.2.0
Reference Date	2026-03-31
Materials Screened	4
PFAS Detected	2
Methodology Hash	sha256:pfas-report-v1.2.0

Databases Used:

- PFAS Substance Registry v1.0 (35 substances)
- PFAS Replacement Scorer v1.0 (30+ candidates)

*This report was generated by KOMPOSOS-III Chemistry Engine.
Scores are based on published literature data. Pilot testing is recommended before full qualification.*